BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: MRD-10TH 2022 (A)

Retyped faithfully from the original board paper — same questions, no changes.

Section-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** VERSION: A

Note: Note: There are three sections in this paper i.e. A, B & C. Instructions: Please attempt Section-A on the MCQs Answer Sheet only. Only black ball point or marker may be used for shading the relevant circle. Please make sure that there is no cutting, over writing, erasing, or multiple circles shading.

Q1. Choose the correct option i.e. A, B, C, or D.

Q1. Radiation therapy is the treatment of .....

(A) Asthma
(B) Lung cancer
(C) Bronchitis
(D) Emphysema

Q2. The most colourful part of the flower is .....

(A) Sepals
(B) Gynoecium
(C) Petal
(D) Androecium

Q3. Arthritis is defined as inflammation in .....

(A) Joints
(B) Muscle
(C) Bones
(D) Cartilage

Q4. The oral drugs are absorbed in the blood and are first taken to .....

(A) Kidneys
(B) Liver
(C) Lungs
(D) Brain

Q5. Internal fertilization occurs in .....

(A) Fish
(B) Amphibians
(C) Aquatic animals
(D) Mammals

Q6. Genes are short segment of .....

(A) Proteins (B) DNA

(C) Carbohydrates (D) Lipids

Q7. Enzymes which are used to cut the DNA in specified sequences is .....

(A) Ligase (B) Protease

(C) DNAase (D) Endonuclease

Q8. Respiration at cellular level is known as .....

(A) Breathing (B) Ventilation

(C) Oxidation (D) Inhalation

Q9. Which of the following acts as exocrine as well as endocrine gland?

(A) Pituitary (B) Thyroid

(C) Adrenal (D) Pancreas

Q10. An ecological pyramid refers to the .....

(A) Pyramid of energy (B) Pyramid of number

(C) Pyramid of biomass (D) All of these

Q11. About 70% kidney stones are formed by .....

(A) Uric acid (B) Calcium oxalate

(C) Vitamins (D) Infections

Q12. Energy flow through an eco-system is always .....

(A) Unidirectional (B) Cyclic

(C) Reversible (D) Multidirectional

Section-B

Time Allowed: 2:40 Hours | **Maximum Marks:** 32

Q2. Write short answers to any eight (8) of the following parts. All parts carry equal marks.

(i) Write a note on hydrophytes plants.

(ii) Differentiate between internal and external fertilization.

(iii) How kidney play role as homeostatic organ?

(iv) State any four functions of bones.

(v) Describe the role of gene in protein synthesis.

(vi) What are the biotic components of ecosystem?

(vii) State the function of cerebrum.

(viii) Name any four sources of water pollution.

(ix) Differentiate between antibiotics and vaccines.

(x) Describe the mechanism of breathing.

(xi) Write a note on "Epilepsy".

Section-C

Maximum Marks: 21

> Note: Answer any three (3) questions. All questions carry equal marks.

Q3. (Marks 7) Write a detailed note on Mendel's Law of independent assortment with example.

Q4.

(a) [4] Discuss any two disorder of the human eye.

(b) [3] Describe the structure of neuron.

Q5.

(a) [4] Define biogeochemical cycle. Discuss nitrogen cycle.

(b) [3] Write a note on adrenal gland.

Q6.

(a) [4] Define air pollution and explain its causes.

(b) [3] Discuss the basic techniques of genetic engineering.